

Hey, everybody. Welcome to your chapter 22 review. This is an important chapter for fitness professionals because exercise modalities are ultimately the vehicle that helps your clients get to the destination of achieving their goals.

Up until this point, we have talked about different exercise components, training systems and concepts, and the OPT model training framework. Now let's dive into the exercise types and techniques that you will use with your clients to stimulate their body to adapt and change.

Change is inevitable, especially in a field like fitness, where scientific methods and consumer-focused products fight for center stage. On one hand, you'll constantly have new peer-reviewed studies and cool innovations to consider. Then, on the other hand, you have a competitive marketplace of people with evolving needs and tastes.

To keep up with it all, you'll need to stay on top of the world of exercise modalities. But fear not. NASM has you covered with great strategies and some universal guidelines to keep you on track.

Basically speaking, exercise modalities are tools that are designed to enhance an exercise or movement to create a desired outcome. Some enhance resistance training, such as free weights, cable machines, and sandbags, to name a few. Others enhance balance, coordination, and joint stability like proprioceptive BOSU balls and the Terra-Core.

And of course, everybody loves a good gadget. There are tons of popular trackers, heart rate monitors, and apps for nutrition and fitness. These provide additional means of accountability, motivation, consistency, and safety. They might also give you, as a CPT, a more effective method of helping your clients achieve their fitness goals.

At the end of the day, it's going to be your responsibility to be well-versed in as many fitness modalities as possible. By having a better understanding of what's out there, you'll be able to design more versatile, efficient, effective, and functional programs.

It's also important to consider your client assessments by keeping the needs, capabilities, and limitations of each client in mind. You can maintain safety when deciding which training modalities may be the best to use and when to integrate them into a program.

By the end of this chapter, you should be able to categorize common exercise modalities and their uses, describe the use of different exercise modalities with clients for specific goals, and determine appropriate uses for various fitness trackers.

While new fitness modalities can offer numerous benefits, there are a few precautions to consider as well. First, never compromise form over function, regardless of the modality. Sacrificing proper exercise technique is never recommended. Only introduce new modalities or advanced forms of exercise after clients have exhibited competency in basic movement patterns.

Additionally, always use caution when training on unstable surfaces. And finally, it's important to recognize the limitations of fitness apps as they are not always 100% accurate. Miscalculations could lead to overexertion, and some movements may not be appropriate for clients with chronic conditions.

That said, let's take a further look into modality options for exercise. Selectorized strength training machines are popular in fitness facilities and are often the first choice for the novice exerciser, deconditioned clients, and older adults, as they offer a safer alternative to free weights and do not require a spotter.

During initial stages of training, they can also offer a less intimidating environment. However, it's important to note that most selectorized machines require less activation of the core musculature than free weights and rarely allow for multi-joint movements.

They are often regarded as inferior to free weights for improving core stability and muscular coordination, because they offer artificial support instead of requiring the user to support themselves with their core musculature.

Moreover, not all strength-training machines are designed to fit all body types, which can also limit the effectiveness of the exercise and possibly create more stress to joints. And lastly, they primarily work on movements occurring in only one plane of motion, as opposed to challenging the body in multiple planes.

Free weight exercises, such as those with dumbbells and barbells, can be used by a variety of populations to meet fitness, wellness, and sports performance goals. They allow exercises to be performed through a greater ROM and in all planes of motion.

All phases of the OPT model can effectively employ free weights. Just keep in mind they can be intimidating for some clients as well as have a learning curve. Overall, free weights can be used in a variety of ways to help achieve many goals by virtually all populations.

Moving on-- cable machines, elastic resistance, and the TRX Rip Trainer provide multidirectional load resistance, similar to free weights, and have been proven to be beneficial in developing core strength. So those are valuable options to include in programs as well.

One thing to keep in mind, however, is when you're using cable machines with clients, it's important to match the cable's resistance to the muscle's natural line of pull. Also noteworthy-- elastic bands and tubing can enable clients to perform resisted exercises that mimic functional and sport-specific movements, like a golf swing or tennis forehand.

Elastic resistance modalities are portable and relatively inexpensive, though they may not be most ideal when trying to develop high levels of strength and muscular hypertrophy.

Medicine balls are one of the oldest means of resistance training, dating back nearly 3,000 years ago. They are balls that come in an assortment of weights, sizes, and materials. They can be used like other resistance implements to add load or instability to an exercise.

Additionally, they can be used with a variety of populations as part of a program to increase muscular strength, endurance, and power, or in some cases, to help rehabilitate from injury. Medicine balls are extremely versatile, but make sure to assess your client and the training environment before implementing programs in which they are thrown or caught.

A kettlebell is a flat-bottomed cast-iron ball with a handle. It's different from a dumbbell, barbell, or medicine ball in that the center of mass is away from the handle, which may require more strength and coordination to maneuver.

It will also engage increased recruitment from stabilizers and prime movers simultaneously during particular movements. And a lot of kettlebell exercises involve multiple joint motions and muscle groups.

Suspended bodyweight training is an innovative approach to bodyweight fitness training. It uses a system of ropes and webbing that allows the user to work against their own body weight while performing various movements and exercises. These exercises do not require an additional load like a dumbbell or strength-training machine.

Common suspended exercises include push-ups, rows, and single-leg squats, to name a few. Suspended bodyweight exercises become a powerful way to teach proper movement patterns, enhance multi-planar stability and core strength, and gain metabolic benefits.

Sandbags provide another unique and versatile way to train your clients. They are designed to be carried, lifted, thrown, and pulled. Most fitness-type sandbags come with several handles to easily change grips. Unlike barbells, dumbbells, and selectorized machines, the sand within the bag is constantly shifting, which gives the benefit of continuous instability.

ViPR is an acronym for Vitality, Performance, and Reconditioning. The ViPR product is a cylindrical tube constructed of hardened rubber with ergonomically placed handles and openings on both ends. It's designed to be dragged, rolled, tossed, lifted, pulled, and carried.

This design enables the ability to perform multi-directional, full-body exercises with external load resistance. This is also known as loaded movement training.

It's amazing to think about how so many of our modern fitness tools are just basic things that we've reimagined for an exercise perspective. Battle ropes are a great example. They're typically made of heavy-duty nylon and come in a variety of lengths and thickness to allow for low-impact activities.

Then we get into balance or proprioceptive modalities. Their primary use is to increase the demand for stability in an exercise. They can be effective for optimizing movement patterns and improving posture, joint stability, balance, and performance.

But keep in mind that they shouldn't be used to perform maximal or near maximal lifts for safety reasons. There's also an inherent risk of falling, so ensure when using proprioceptive modalities, exercises are always performed with correct form and good posture to maintain technique and safety throughout.

Stability balls, also called Swiss balls, are used in a variety of training facilities with a wide range of populations. Their primary use is to increase the demand for stability in an exercise. But they can also be used for added comfort and support or to reinforce proper posture during certain movements, such as squatting.

Stability balls are best used with clients who demonstrate a need for increased overload of stability. As with any exercise, a proper progression from stable to unstable surfaces should be observed.

Next, the BOSU ball is an inflated rubber hemisphere attached to a solid plastic base. It looks like a stability ball cut in half. When the flat side is down, the dome offers a surface similar to a stability ball, yet stable enough to stand on.

And when the flat side is up, it provides a platform on which the hands can be placed to perform upper-body exercises. Training with a BOSU ball offers the ability to increase the intensity of an exercise by decreasing stability.

The Terra-Core is made of an inflatable rubber bladder and hard-surfaced backing, not unlike a BOSU ball, but is rectangular in shape instead of round, larger in surface area, and offers handles and sidebars, which permits additional versatility and variety during exercise.

Unlike stability balls, it's generally safe to perform a lot of your resistance training exercises, like a Dumbbell Chest Press, while lying on the Terra-Core.

Finally, we return to those fun techie devices and apps. Fitness trackers are primarily electronic wearable devices that enable a user to track their activity levels. They come in the form of watches, bands, rings, heart rate monitors, pedometers, and more.

They tend to be easy to use and provide intrinsic motivation, which are key factors for continued use. However, while they can absolutely help increase motivation, accountability, and athletic performance, keep in mind they aren't always 100% accurate. Regardless, these devices are still good additions to any well-designed program, especially to monitor and track exercise effort.

There are going to be pros and cons with every modality. Selection of the appropriate combination of modalities will come down to what is appropriate for your client, as well as what aligns with their needs and wants.

Use your education and experience to pick and choose the right combination of tools to build a cohesive program and keep your eye on new research and development in this field. It will give you an edge in program design and maybe even bolster your engagement level in what science is presenting to us.

If you have a good understanding of the wide variety of modalities available, it'll make it easier to adapt and be flexible in your programming. If you're working in a busy commercial fitness facility, for example, and the piece of equipment you'd intended to use isn't available, then understanding other modalities that could be used in place of the original will really come in handy.

Additionally, if you have a client that is partial to a particular exercise modality, incorporating that into their training may enhance their enjoyment and adherence. Until next time, take care, and awesome job coming this far in your training.